

8 Day Report

Task: Azure ARM Deployment & Disk Configuration

Intern Name: Syed Fuzail

Day: 8

Technology Used: Microsoft Azure (ARM Templates, VM, Managed Disks, Custom Script Extension)

Resource Group: syedfuzail2009-rg

Objective of the Day

The objective of Day 8 was:

1. Deploy a Windows Virtual Machine using Azure ARM Template
 2. Configure networking components (VNet, Subnet, NSG, Public IP, NIC)
 3. Install IIS using Custom Script Extension
 4. Deploy a custom HTML page
 5. Create and attach a new managed data disk to the existing VM
 6. Verify disk configuration inside the virtual machine
-

Task 1 – Deploy Windows VM using ARM Template

◆ Infrastructure Components Created

- Virtual Network (VNet)
- Subnet

- Network Security Group (NSG)
 - Public IP Address
 - Network Interface Card (NIC)
 - Windows 11 Virtual Machine
 - Custom Script Extension
-

◆ Network Configuration

Two inbound security rules were configured in NSG:

Port	Purpose
3389	RDP Access
80	HTTP Access (IIS)

This allowed:

- Remote Desktop access to the VM
 - Web access to the hosted HTML page
-

◆ Virtual Machine Configuration

- OS: Windows 11 Pro (Latest)
 - VM Size: Based on allowed SKU in subscription
 - OS Disk: Default Managed Disk
 - Authentication: Username & Password
-

◆ Custom Script Extension

A PowerShell script was uploaded to Azure Storage and executed using:

`Microsoft.Compute/virtualMachines/extensions`

Script Actions:

- Installed IIS (Web-Server Role)

Created custom HTML page in:

`C:\inetpub\wwwroot\index.html`

- - Restarted IIS service
 - Verified web hosting functionality
-

◆ Output Verification

After deployment:

1. Public IP address was retrieved

Accessed in browser:

`http://<Public-IP>`

- 2.
 3. Successfully displayed custom IIS page
-



Task 2 – Create and Attach a New Managed Data Disk

◆ Objective

To create a separate managed disk and attach it to the existing Windows VM.

◆ Disk Configuration

- Disk Type: Standard SSD (StandardSSD_LRS)
- Size: 64 GB
- Disk Name: `armDataDisk01`
- LUN: 0

◆ ARM Implementation

The template included:

`Microsoft.Compute/disks`

And attached to VM under:

`storageProfile → dataDisks`

◆ Inside VM Configuration

After deployment:

1. Connected to VM using RDP

Opened Disk Management:

`diskmgmt.msc`

- 2.
3. Initialized new disk
4. Created new simple volume
5. Formatted with NTFS

6. Disk appeared as D: drive
-

Challenges Faced

During implementation, several issues were encountered:

1 VM SKU Not Available

- Some VM sizes were restricted in the subscription
- Resolved by checking available SKUs

2 Authorization Error (RBAC)

Error:

```
AuthorizationFailed  
Microsoft.Resources/deployments/validate/action
```

Cause:

- Insufficient permissions (Contributor role required)

Resolution:

- Verified IAM permissions
- Adjusted deployment method

3 Deployment Conflicts

- Existing resources caused name conflicts
 - Resolved by modifying resource names
-



Key Learnings

- Understanding Azure ARM template structure
 - Managing dependencies using `dependsOn`
 - Working with managed disks
 - Using Custom Script Extension
 - Handling RBAC permission errors
 - Troubleshooting SKU capacity issues
 - Managing LUN configuration for data disks
-



Security Considerations

- NSG rules were restricted to required ports only
 - Standard SSD used for cost efficiency
 - Proper authentication configured
 - Deployment controlled via ARM templates
-



Final Outcome

By the end of Day 8:

- ✓ Windows VM successfully deployed
 - ✓ IIS installed automatically
 - ✓ Custom HTML page hosted
 - ✓ Managed data disk created and attached
 - ✓ Disk initialized and formatted
 - ✓ Full infrastructure deployed using Infrastructure as Code
-



Conclusion

Day 8 focused on practical implementation of Azure Infrastructure as Code using ARM Templates.

This session enhanced understanding of:

- Cloud Infrastructure Automation
- VM configuration
- Disk management
- Network security configuration
- Azure troubleshooting techniques

The objectives of the day were successfully achieved.

1 ARM Template – Create Windows Virtual Machine

 Save as: **vm-template.json**

```
{
  "$schema":
    "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "vmName": { "type": "string", "defaultValue": "armVM01" },
    "adminUsername": { "type": "string", "defaultValue": "azureuser" },
    "adminPassword": { "type": "secureString" },
    "location": { "type": "string", "defaultValue": "[resourceGroup().location]" }
```

```
},  
"resources": [  
  {  
    "type": "Microsoft.Compute/virtualMachines",  
    "apiVersion": "2023-03-01",  
    "name": "[parameters('vmName')]",  
    "location": "[parameters('location')]",  
    "properties": {  
      "hardwareProfile": {  
        "vmSize": "Standard_B1s"  
      },  
      "osProfile": {  
        "computerName": "[parameters('vmName')]",  
        "adminUsername": "[parameters('adminUsername')]",  
        "adminPassword": "[parameters('adminPassword')]"  
      },  
      "storageProfile": {  
        "imageReference": {  
          "publisher": "MicrosoftWindowsServer",  
          "offer": "WindowsServer",  
          "sku": "2022-Datacenter",  
          "version": "latest"  
        },  
        "osDisk": {
```



```
        "createOption": "FromImage"
      }
    },
    "networkProfile": {
      "networkInterfaces": []
    }
  }
}
]
```

⚠ Note: This VM template assumes networking already exists.
If you need full networking included, tell me.

2 ARM Template – Create Storage Account

📁 Save as: **storage-template.json**

```
{
  "$schema":
    "https://schema.management.azure.com/schemas/2019-04-01/deploymentTe
    mplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "storageAccountName": {
      "type": "string"
```

```

    },
    "location": {
      "type": "string",
      "defaultValue": "[resourceGroup().location]"
    }
  },
  "resources": [
    {
      "type": "Microsoft.Storage/storageAccounts",
      "apiVersion": "2023-01-01",
      "name": "[parameters('storageAccountName')]",
      "location": "[parameters('location')]",
      "sku": {
        "name": "Standard_LRS"
      },
      "kind": "StorageV2",
      "properties": {}
    }
  ]
}

```

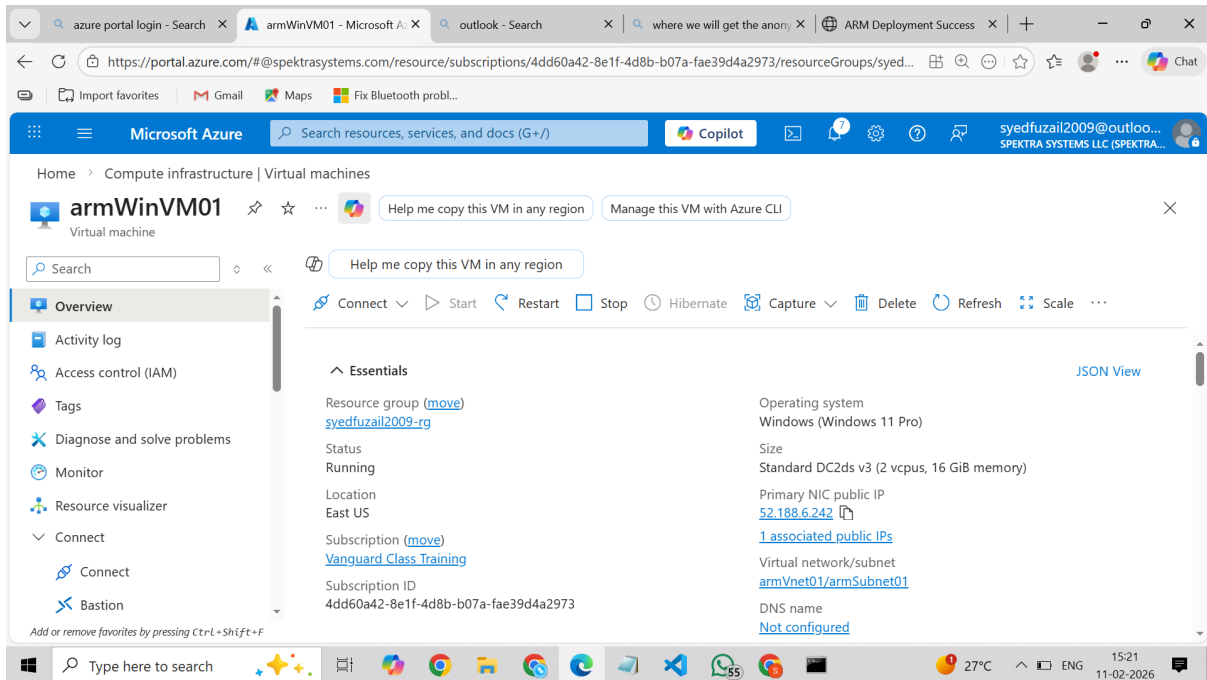
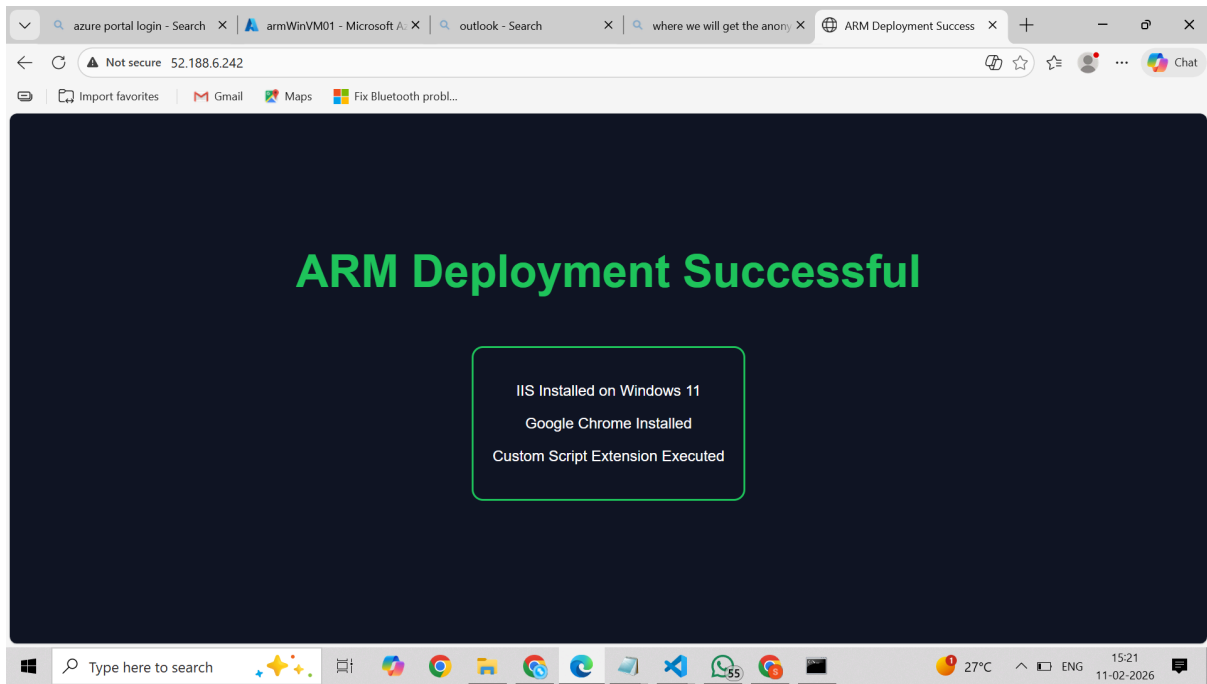
- ◆ Storage type: General Purpose v2
 - ◆ Replication: Locally Redundant (LRS)
-

3 ARM Template – Create Azure Container Registry (ACR)

 Save as: **acr-template.json**

```
{
  "$schema":
    "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "acrName": {
      "type": "string"
    },
    "location": {
      "type": "string",
      "defaultValue": "[resourceGroup().location]"
    }
  },
  "resources": [
    {
      "type": "Microsoft.ContainerRegistry/registries",
      "apiVersion": "2023-01-01-preview",
      "name": "[parameters('acrName')]",
      "location": "[parameters('location')]",
      "sku": {
```

```
      "name": "Basic"
    },
    "properties": {
      "adminUserEnabled": true
    }
  ]
}
```



Home

syedfuzail2009-rg

Resource group

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Overview

- Activity log
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Name	Type	Location
armNic01	Network Interface	East US
armNsg01	Network security group	East US
armPublicIp01	Public IP address	East US
armVnet01	Virtual network	East US
armWinVM01	Virtual machine	East US
armWinVM01_OsDisk_1_138a0c06326a48eb970bdb36082	Disk	East US
syed11111	Storage account	East US

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Resource group

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```
graph TD; armVnet01[armVnet01 Virtual network] --- armNic01[armNic01 Network Interface]; armVnet01 --- armNsg01[armNsg01 Network security group]; armVnet01 --- armPublicIp01[armPublicIp01 Public IP address]; armWinVM01[armWinVM01 Virtual machine] --- armVnet01; armWinVM01 --- armNic01; armWinVM01 --- armWinVM01_OsDisk_1_138a0c06326a48eb970bdb36082[armWinVM01_OsDisk_1_138a0c06326a48eb970bdb36082 Disk]; syed11111[syed11111 Storage account] --- armWinVM01; syed11111 --- armWinVM01_OsDisk_1_138a0c06326a48eb970bdb36082;
```

azure portal login - Search X Microsoft.Template-202602 X outlook - Search X where we will get the anony X ARM Deployment Success X + -

https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2F4dd60a42-8e1f-4d8b-b07a-... Import favorites Gmail Maps Fix Bluetooth probl...

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Microsoft.Template-20260211153743 | Overview

Deployment

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✓ Your deployment is complete

Deployment name : Microsoft.Template-20260211153743
Subscription : Vanguard Class Training
Resource group : syedfuzail2009-rg
Start time : 2/11/2026, 3:37:49 PM
Correlation ID : dfd38153-61e5-484b-aa1a-33056fcfad3c

> Deployment details

✓ Next steps

[Go to resource group](#)

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```
C:\Windows\System32\cmd.exe
Tenant: Spektra Systems LLC
Subscription: Vanguard Class Training (4dd60a42-8e1f-4d8b-b07a-fae39d4a2973)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

C:\Users\Lenovo\Desktop\aaam>2
'2' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Lenovo\Desktop\aaam>az deployment group create --resource-group "syedfuzail2009-rg" --template-file mainTemplate.json --parameters parameters.json
{"code": "AuthorizationFailed", "message": "The client 'live.com#syedfuzail2009@outlook.com' with object id '61c29db9-e40d-44b0-b76b-c3529575f7d6' does not have authorization to perform action 'Microsoft.Resources/deployments/validate/action' over scope '/subscriptions/4dd60a42-8e1f-4d8b-b07a-fae39d4a2973/resourcegroups/syedfuzail2009-rg/providers/Microsoft.Resources/deployments/mainTemplate' or the scope is invalid. If access was recently granted, please refresh your credentials."}

C:\Users\Lenovo\Desktop\aaam>

C:\Users\Lenovo\Desktop\aaam>

C:\Users\Lenovo\Desktop\aaam>
```